Progress Report: Systematic Literature Review on the Effect of Prototyping on Identifying Missing and Unclear Requirements

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***Abstract*—This academic progress report synthesizes early empirical and theoretical findings for the topic: "The effect of prototyping on identifying missing and unclear requirements". Rooted in foundational Requirements Engineering (RE) principles from the IREB/CPRE curriculum and industrial technique maturity evaluations, we critically examine three primary empirical studies. We trace an evolutionary research storyline: establishing multi-dimensional classification models (PAM), quantifying elicitation volume while isolating non-functional requirement (NFR) deficiencies, and rectifying NFR blind spots via verbalized interactive paper prototyping. A systematic cross-study synthesis matrix, critical research gaps, and a future research roadmap are formally presented.**

***Index Terms*—Requirements Engineering, Prototyping, Missing Requirements, Ambiguity Resolution, Usability, SLR.**

I. Introduction

Requirements Engineering (RE) is inherently vulnerable to defects originating from human cognitive and linguistic communication barriers. As conceptualized by Pohl and Rupp in the CPRE/IREB foundational curriculum [1], textual Software Requirements Specifications (SRS) based on natural language suffer from five fundamental transformational effects: nominalization, nouns lacking reference objects, universal quantifiers, incomplete verbs, and incomplete conditions (e.g., specifying "If" scenarios while omitting corresponding "Else" edge cases). Consequently, purely textual specifications predominantly surface conscious requirements (*Satisfiers* in the Kano Model [1]), while obscuring taken-for-granted baseline needs (*Dissatisfiers*) and unspoken innovation opportunities (*Delighters*).

To resolve these representational deficiencies, IREB Principle 4 (*Adequate Change of Documentation Type*) and Principle 5 (*Early Artifact Construction*) mandate shifting representation modalities from abstract prose to concrete, interactive artifacts [1]. Prototyping acts as a cognitive catalyst, providing direct sensory feedback to validate users' internal mental models and exposing unspoken requirements [1]. Furthermore, the systematic literature review by Didar et al. [2] confirms that prototyping is an industrially mature elicitation technique with over a decade of empirical validation, ranking among the top three methods globally. Didar et al. [2] highlight that prototyping is predominantly deployed within hybrid configurations (accounting for 22% of multi-technique studies), most notably coupled with Scenarios to foster continuous bidirectional alignment and prevent requirements defects.

II. Review Methodology

This progress report adopts a systematic literature review methodology aligned with formal empirical software engineering standards. An empirical data extraction matrix was structured to systematically capture: empirical context, prototyping taxonomy (aligned with the PAM framework), comparative baseline performance against textual methods (RQ6.2), requirement categorization impacts (RQ6.3), and statistical evidence (p-values, effect sizes). Primary studies were selected based on peer-reviewed indexing (IEEE, Springer, Elsevier) providing direct empirical comparisons between prototyping modalities and traditional elicitation channels. To date, full data synthesis has been completed for 2 foundational sources and 3 primary empirical anchor investigations.

III. Thematic Review

Rather than examining papers in isolation, findings are synthesized across three core thematic dimensions:

A. Prototyping Taxonomy and Use (RQ6.1)

Bjarnason et al. [3] formalized empirical prototyping practices via the Prototyping Aspects Model (PAM), structuring implementations along five dimensions: *Purpose* (exploration, alignment, validation), *Scope* (breadth and functional refinement), *Media* (paper sketches to executable source code), *Evaluation Use* (internal/external review protocols), and *Exploration Strategy* (point-based vs. parallel). Didar et al. [2] and Pohl and Rupp [1] further classify lifecycle utilization into Throwaway and Evolutionary modalities, confirming that prototypes achieve maximum efficacy when embedded within scenario-based walkthroughs.

B. Efficacy in Surfacing Missing and Ambiguous Requirements (RQ6.2)

Field evidence from Bjarnason et al. [3] demonstrated that interactive mockups significantly outperform formal specifications in resolving unaligned views and exposing omission defects. In controlled comparisons, Rueda et al. [4] confirmed via Kruskal-Wallis testing that Paper Prototyping yielded the highest total quantity of requirements ($p = 0.02$, effect size =

0.08) compared to JAD and Unstructured Interviews. However, passive paper mockups showed no statistically significant divergence regarding specification completeness ($p = 0.17$) or correctness ($p = 0.59$).

C. Impact by Requirement Category (RQ6.3)

While prototyping consistently accelerates the elicitation of functional requirements and UI structures [4], traditional low-fidelity mockups exhibited an acute deficit in capturing Non-Functional Requirements (NFRs) [4]. Abad et al. [5] rectified this imbalance by evaluating Loud Paper Prototyping (LPP), pairing Wizard-of-Oz interactions with a concurrent think-aloud protocol. LPP achieved statistically significant improvements over traditional meetings across User Interface requirements ($p = 0.0001$) and critical NFRs including usability, learnability, and navigation ($p = 0.001$).

IV. Comparative Analysis and Synthesis

Table I presents the comparative synthesis matrix across the investigated empirical studies.

The synthesis indicates a clear evolutionary progression: theoretical foundations explain why natural language fails; the PAM taxonomy organizes prototype operationalization; comparative trials reveal the volume strengths and NFR vulnerabilities of paper prototypes; and verbalized interaction modalities resolve those NFR vulnerabilities.

V. Research Gaps

Critical gaps identified from the synthesized literature include:

1) **Absence of Direct Semantic Ambiguity Metrics:** Empirical studies predominantly quantify requirement volume or completeness, lacking formal defect-inspection metrics specifically evaluating the reduction of linguistic ambiguity.
2) **Academic vs. Industrial Contextual Divide:** Rigorous comparative trials are heavily concentrated within academic student cohorts, whereas industrial investigations remain predominantly qualitative.
3) **Cost-Fidelity Trade-offs:** Prototyping incurs substantial cost overheads [2], with high-fidelity variants introducing premature commitment risks [3].

VI. Proposed Research Direction

Subsequent investigation phases will focus on:

- **Ambiguity Detection Metrics:** Synthesizing empirical evidence regarding formal ambiguity checklists and interface validation mechanisms.
- **Agile Integration:** Investigating the coupling of low-fidelity prototypes with User Stories and backlog grooming to mitigate late-stage rework.
- **Matrix Expansion:** Extracting data from remaining candidate studies in the 18-paper corpus to complete the comprehensive SLR.

VII. Conclusion

This preliminary progress report confirms that prototyping functions as a vital operationalization of IREB Principle 4, successfully bypassing the linguistic transformations of textual SRS. The synthesis of three core empirical studies demonstrates that while passive prototyping maximizes functional requirement capture, augmented verbalization (think-aloud protocols) is essential to overcome non-functional usability blind spots. The established synthesis matrix provides a rigorous foundation for completing the full systematic review.

References

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TABLE I
CROSS-STUDY COMPARATIVE SYNTHESIS MATRIX

Paper / Reference	**Empirical Sample**	**Prototyping Modality**	**Key Evidence on Missing / Unclear Reqs**	**Critical Research Gaps**
Bjarnason et al. (2023) [3]	11 startups, 43 instances	Variable fidelity; formal PAM categorization	Visual artifacts resolve unaligned views; interactive walkthroughs reveal missed requirements	Premature idea cementing; premature release pressure
Rueda et al. (2020) [4]	4 replications, 167 students, 30 teams	Low-fi paper mockups vs. JAD vs. Interviews	Significant increase in total volume ($p = 0.02$); excels at functional/UI elicitation	Deficient in eliciting NFRs; non-significant completeness ($p = 0.17$)
Abad et al. (2018) [5]	183 students, 50 mobile app teams	Wizard of Oz paper mockups + Think-Aloud	Significant capture of UI ($p < 0.001$) and Usability NFRs ($p = 0.001$); 20% added missing features	Interaction slowdown; potential facilitator simulation bias